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CABLE MANAGEMENT DEVICE

Abstract

A cable management device for mounting to a telecommunications fixture includes an outer barrel disposed over an inner barrel, one of the outer barrel and the inner barrel defining a plurality of discrete detents positioned in a stacked arrangement axially along an length thereof, and the other of the outer barrel and the inner barrel defining at least one flexible cantilever arm defining a tab configured to lock into a selected one of the detents for allowing adjustment of a length of the cable management device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/668,541, filed on May 20, 2024; which is a continuation of U.S. patent application Ser. No. 18/323,673, filed on May 25, 2023, now U.S. Pat. No. 11,988,884; which is a continuation of U.S. patent application Ser. No. 17/586,518, filed on Jan. 27, 2022, now U.S. Pat. No. 11,668,888; which is a continuation of U.S. patent application Ser. No. 16/902,770, filed on Jun. 16, 2020, now U.S. Pat. No. 11,262,518; which is a continuation of U.S. patent application Ser. No. 16/324,191, filed on Feb. 8, 2019, now U.S. Pat. No. 10,690,874; which is a National Stage Application of PCT/US2017/045703, filed on Aug. 7, 2017; which claims the benefit of U.S. Patent Application Ser. No. 62/372,123, filed on Aug. 8, 2016, the disclosures of which are incorporated herein by reference in their entireties. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

TECHNICAL FIELD

[0002] The present disclosure relates generally to devices for enhancing cable management of a telecommunications system. More particularly, the present disclosure relates to a cable management assembly including a cable management spool used on a telecommunications panel.

BACKGROUND

[0003] Telecommunications systems commonly include cables (e.g., fiber optic cables, or copper twisted pair cables) for interconnecting pieces of telecommunications equipment.

Telecommunications racks typically hold a variety of different pieces of telecommunications equipment. Often thousands of cables are used to interconnect the various pieces of telecommunications equipment mounted on the racks.

[0004] Because of the large number of cables associated with telecommunications equipment, cable management is crucial. Cable management involves efficiently routing cables to minimize the space occupied by such cables. Cable management also involves routing cables in an orderly manner so as to reduce the likelihood of cable tangling. Ease of cable organization is also a factor related to effective cable management.

[0005] Cable management is also important in preventing damage to the cables. Unnecessary or excessive displacement of fiber optic cables, for example, is undesirable. As the fiber optic cables are displaced, they are subject to bending and other forces. Bending of the fibers can cause attenuation and loss of signal strength. As the fiber bends, the fiber can also break, resulting in a loss of transmission through the fiber.

[0006] According to one example type of cable management, cable spools may extend from a surface in columns, rows, or other configurations. One or more fiber optic cables may be hung over or wrapped around a curved surface of one or more of these spools to take up the slack length of the fiber optic cable.

[0007] In general, cable management improvement has been sought, generally to efficiently and effectively manage cables by providing system adaptability, and ease of cable organization, especially in such spool-based systems.

SUMMARY

[0008] One aspect of the present disclosure relates to a telecommunications system including a cable management assembly including a panel having a plurality of shaped apertures and a plurality of cable management devices in the form of spools for mounting to the panel. Each cable management device includes attachment features corresponding to the shaped apertures that secure the cable management devices to the panel at selected horizontal and vertical locations. According to one inventive aspect, the cable management devices in the form of spools may include features for adjusting or varying the axial lengths thereof for equipment clearance or space-adaptability.

[0009] According to another aspect, the disclosure relates to a cable management device in the form of a cable spool defining an outer barrel that is disposed over an inner barrel, both having a generally round configuration for providing cable management with bend radius protection. One of the outer barrel and the inner barrel includes a plurality of discretely spaced locking detents along the length thereof, and the other of the outer barrel and the inner barrel includes a flexible arm with a tab that can lock within one of the detents. Relative rotation between the outer barrel and the inner barrel allows the tab to enter or exit the detent. For example, after a desired length for the spool is determined, relative rotational movement between the outer barrel and the inner barrel allows the tab to lock within a detent that corresponds to the desired length, thus providing adjustability to the length of the overall cable management spool.

[0010] Another aspect of the present disclosure relates to a cable management assembly comprising a telecommunications panel including a plurality of shaped apertures arranged in an array of columns and rows and a plurality of cable management devices having attachment structures corresponding to the shaped apertures for removably mounting the cable management devices to the panel at selected vertical and horizontal locations, wherein each cable management device defines an outer barrel disposed over an inner barrel, one of the outer barrel and the inner barrel defining a plurality of discrete detents positioned in a stacked arrangement axially along a length thereof, and the other of the outer barrel and the inner barrel defining at least one flexible cantilever arm defining a tab configured to lock into a selected one of the detents for allowing adjustment of a length of the cable management device.

[0011] According to another aspect, the present disclosure relates to a cable management device for mounting to a telecommunications fixture, the cable management device comprising an outer barrel disposed over an inner barrel, one of the outer barrel and the inner barrel defining a plurality of discrete detents positioned in a stacked arrangement axially along a length thereof, and the other of the outer barrel and the inner barrel defining at least one flexible cantilever arm defining a tab configured to lock into a selected one of the detents for allowing adjustment of a length of the cable management device.

[0012] According to yet another aspect, the present disclosure relates to a method of using a cable management device, the method comprising adjusting a length of the cable management device by relatively sliding an outer barrel and an inner barrel of the cable management device to select the length of the cable management device and relatively rotating the outer barrel and the inner barrel and locking a tab of a flexible cantilever arm of one of the outer barrel and the inner barrel into a detent of a plurality of discrete detents positioned axially along a length of the other of the outer barrel and the inner barrel.

[0013] A variety of inventive aspects are set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practicing various aspects of the disclosure. The aspects of the disclosure may relate to individual features as well as combinations of features. It is to be understood that both the foregoing general description and the following

detailed description are exemplary and explanatory only, and are not restrictive of the claimed inventions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. **1** illustrates a telecommunications system including a cable management assembly formed from a telecommunications panel and at least one cable management device having features that are examples of inventive aspects in accordance with the present disclosure removably mounted to the panel, the cable management assembly shown from an outer perspective view.

[0015] FIG. **2** illustrates the cable management assembly of FIG. **1** from an inner perspective view.

[0016] FIG. **3** illustrates a perspective view of the cable management device of FIGS. **1-2** from a free end thereof.

[0017] FIG. **4** illustrates a perspective view of the cable management device of FIGS. **1-2** from a mounting end thereof.

[0018] FIG. **5** illustrates the mounting of the cable management device of FIGS. **1-4** to the telecommunications panel shown in FIGS. **1-2**.

[0019] FIG. **6** illustrates a close-up view of the cable management device mounted to the panel of FIG. **2**.

[0020] FIG. **7** illustrates a perspective view of an outer barrel of the cable management device of FIGS. **1-6**.

[0021] FIG. **8** illustrates a perspective view of an inner barrel of the cable management device of FIGS. **1-6**.

[0022] FIG. **9** is a partial cross-sectional view bisecting the cable management device of FIGS. **1-6** showing the outer and inner barrels thereof in a locked position.

[0023] FIG. **10** is a partial lateral cross-sectional view of the cable management device of FIGS. **1-6** showing the outer and inner barrels thereof in a locked position.

[0024] FIG. **11** is a partial cross-sectional view similar to that of FIG. **10**, showing the outer and inner barrels thereof in an unlocked position.

[0025] FIG. **12** illustrates another embodiment of a telecommunications system including a cable management assembly formed from a telecommunications panel and a second example of a cable management device having features that are examples of inventive aspects in accordance with the present disclosure removably mounted to the panel, the cable management assembly shown from an outer perspective view.

[0026] FIG. **13** illustrates the cable management assembly of FIG. **12** from an inner perspective view.

[0027] FIG. **14** illustrates a perspective view of the cable management device of FIGS. **12-13** from a free end thereof.

[0028] FIG. **15** illustrates a perspective view of the cable management device of FIGS. **12-13** from a mounting end thereof.

DETAILED DESCRIPTION

[0029] Reference will now be made in detail to exemplary aspects of the present disclosure that are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0030] One embodiment of a telecommunications system **10** for managing cables interconnected to telecommunications components is illustrated in FIGS. **1-11**. The telecommunication system **10** includes a cable management assembly **12** in accordance with the principles disclosed. The cable management assembly **12** is illustrated specifically in FIGS. **1, 2, 5, and 6** and includes a telecommunications panel **14** having a plurality of shaped apertures **16** and a plurality of cable

management devices **18**. In the depicted embodiment, the cable management devices **18** are provided in the form of spools **20**. The panel **14** may also be referred to as a back plane or a riser. [0031] Still referring to FIGS. **1-11**, each cable management device **18** includes attachment features **22** corresponding to the shaped apertures **16** that secure the cable management devices **18** to the panel **14** at selected horizontal and vertical locations. The discrete apertures/openings **16** of the panel **14** and the cable management devices **18** are configured and sized so that a user can organize and arrange cables at various locations along the panel **14**. As will be discussed in further detail, according to one inventive aspect, the cable management devices **18** may include features for adjusting or varying the axial lengths **L** thereof for equipment clearance or space-adaptability.

[0032] One feature of the present disclosure relates to cost reduction associated with fixed or custom-made cable management systems. The present cable management assembly **12** permits a user to configure the system **10** to a particular need. For example, in accordance with the principles disclosed herein, a user has the flexibility to position any number of the cable management devices **18** at a plurality of different locations on the panel **14** of the cable management assembly **12**. In particular, the preferred panel **14** is constructed to provide a plurality of placement coordinates (i.e., vertical and horizontal locations) from which a user may choose to position the cable management devices **18**. Thus, with this design, a user is not limited to a pre-made, fixed cable management system, which may not fit a particular need. Similarly, with this design, the user reduces or eliminates modification and customizing costs associated with reworking a system to fit the particular need.

[0033] Another feature of the present system **10** relates to the system's adaptability and ease of use. In one embodiment, the cable management devices **18** are detachably secured to the riser or panel **14**. This feature permits a user to modify a system's cable management strategy to accommodate growth or changes needed in a quickly changing industry.

[0034] Still referring to FIGS. **1-11** in general, each cable management device **18** in the form of a spool **20** is configured to facilitate adding one or more fiber optic cables or optical fibers to a densely packed fiber storage region. The cable management device **18**, in the given embodiment, is formed from an outer barrel **24** that is disposed over an inner barrel **26**. The barrels **24, 26**, cooperatively, define a storage surface **28** that extends between a first end **30** and a second end **32** of the spool **20**.

[0035] In the example shown, both the outer barrel **24** and the inner barrel **26** forming the management device **18** define a circle-shaped lateral cross-dimensional profile **34**.

[0036] At the first end **30** of the spool **20**, attached to the inner barrel **26**, is defined a latch plate **36**. And, at the second end **32** of the spool **20**, extending from the outer barrel **24**, is defined a flange **38** for acting as a cable retainer to facilitate maintaining the fiber optic cables on the storage surface **28**.

[0037] At the first end **30** of the spool **20**, the latch plate **36** includes features **22** for latching the spool **20** to a piece of telecommunications equipment such as the panel **14** shown in FIGS. **1, 2, 5**, and **6**. The latch plate **36** includes a catch **40**, an abutment member **42** at an upper portion thereof, and a snap-in latch member **44** at a lower portion thereof. The latch member **44** defines a latch tab **46**.

[0038] The upper and the lower features **22** of the latch plate **36** are configured to cooperate with corresponding shaped apertures **16** on the panel **14** for mounting the spools **20** to desired positions on the panel **14** as discussed previously. In general, the cable management device **18** is designed to be secured to the panel **14** by insertion into the panel **14** from one side, without the use of tools or extra fastening components. As shown in detail in FIGS. **5-6**, in mounting a spool **20** to a panel **14**, the upper catch **40** is first inserted through an aperture **16** in the panel **14**. Once a retention portion **48** of the catch **40** is hooked against the panel **14**, the latch plate **36**, and thus the entire spool **20**, is pivoted down to insert the latch member **44** through another lower aperture **16** that is spaced apart from the upper aperture **16** of the panel **14** along the same column. The latch member **44** is flexibly

inserted until the latch tab **46** clears the panel wall **50** and the spool **20** is latched into place. The abutment member **42** at the upper portion of the latch plate **36** is configured to contact a portion of the panel wall **50** defining a lower edge **52** of the upper aperture **16** and provide further stability to the latched spool **20**.

[0039] In releasing the spool **20** from the panel **14** and disassembling the cable management assembly **12**, the portion of the latch member **44** defining the latch tab **46** is simply flexed upwardly until the tab **46** clears the aperture **16** and the spool **20** is pivoted away from the panel **14**. A final downward movement of the spool **20** in unhooking the upper catch **40** releases the spool **20** from the panel **14**.

[0040] Now referring to FIGS. **7-11** specifically, as discussed above and as will be discussed in further detail below, the outer barrel **24** and the inner barrel **26** include features that provide adjustability of the length **L** of the overall spool **20**. Allowing adjustability of the length **L** of the spool **20** facilitates clearance for other equipment that may be positioned adjacent the panel **14**.

[0041] In the example shown, the outer barrel **24** defines a plurality of discretely positioned detents **54** at the inner surface **56** thereof extending axially along the outer barrel **24**. It should be noted that two sets of detents **54** may be positioned at the inner surface **56**, on opposing sides thereof.

[0042] In the example shown, the inner barrel **26** defines a flexible cantilever arm **58** on opposing sides of the outer surface **60** thereof. The flexible cantilever arms **58** define tabs **62** that are configured to project radially outwardly from the outer surface **60** of the inner barrel **26**. As will be discussed, the tabs **62** are configured to mate with and lock into the detents **54** provided on the inner surface **56** of the outer barrel **24** in positionally locking the outer barrel **24** to the inner barrel **26**. When the tabs **62** are not within the detents **54**, the outer barrel **24** can be slidably moved in a telescoping fashion with respect to the inner barrel **26** in adjusting the length **L** of the spool **20**. And, once the desired length **L** is set and the tabs **62** are in alignment with the detents **54** that correspond to the selected length **L** of the spool **20**, relative rotation of the outer barrel **24** over the inner barrel **26** moves the tabs **62** into the detents **54** to lock the outer barrel **24** with respect to the inner barrel **26**.

[0043] If further adjustment is needed, relative rotation between the outer barrel **24** and the inner barrel **26** once again allows a tab **62** or a pair of tabs **62** to exit a detent **54** or a pair of detents **54**. After a desired length **L** for the spool **20** is determined, relative rotational movement between the outer barrel **24** and the inner barrel **26** once again allows the tab **62** to lock within another of the detents **54** that corresponds to the desired length **L**. Thus, the telescoping slidable movement and the rotational locking movement between the outer and inner barrels **24, 26** provide adjustability to the length **L** of the overall cable management spool **20**.

[0044] FIGS. **9-10** illustrate a locked position for the spool **20** with one of the tabs **62** of the flexible cantilever arms **58** within one of the detents **54**. FIG. **11** illustrates an unlocked position for the spool **20**, wherein a tab **62** or a pair of tabs **62** are outside of a given detent **54** or pair of detents **54**. In the unlocked position shown in FIG. **11**, the outer barrel **24** is able to telescopically slide with respect to the inner barrel **26** so as to provide adjustment for the length **L** of the spool **20**.

[0045] It should be noted that the tabs **62** on the flexible cantilever arms **58** may include tapered surfaces **64** to help with slidable insertion into and removal from the detents **54**. As shown in the cross-sectional views in FIGS. **10** and **11**, the detents **54** may also define tapered entry/exit surfaces **66** to facilitate insertion and removal of the tabs **62**.

[0046] It will be appreciated that, even though in the depicted embodiment, the outer barre **124** is the structure that is configured with the detents **54** and the inner barrel **26** is the structure that is configured with the cooperating cantilever arms **58**, in other embodiments, these structures may be reversed.

[0047] FIGS. **12-15** illustrate a cable management assembly **112** including a panel **14** and a cable management device **118** that has features similar to those described above with respect to the device **18** of FIGS. **1-11**. In the device **118** of FIGS. **12-15**, both the outer and the inner barrels **124**,

126 of the device **118** are depicted as having been molded from two semi-circular pieces **119** that are fitted together with a snap-fit interlock to form the circular lateral cross-dimensional profile **34**. In the embodiment of the spool **20** illustrated in FIGS. **1-11**, the outer and the inner barrels **24**, **26** are illustrated as structures that are molded as a single piece. Otherwise, the devices **18**, **118** of FIGS. **1-11** and **12-15** function in a similar manner.

[0048] The above specification provides a complete description of the telecommunications systems including the cable management assemblies and methods of use. Since many embodiments of the disclosure can be made without departing from the spirit and scope of the inventive aspects, the inventive aspects reside in the claims hereinafter appended.

Claims

1-20. (canceled)

21. A cable management assembly comprising: a telecommunications frame defining a plurality of apertures; and a plurality of cable management devices mounted to the telecommunications frame, the cable management devices having attachment structures corresponding to the apertures defined by the telecommunications frame for removably mounting the cable management devices to the telecommunications frame at selected locations on the telecommunications frame, wherein each cable management device defines a first cylindrical structure defining a wall with a circular lateral cross-dimensional profile, wherein each cable management device also defines a second structure defining a wall with a circular lateral cross-dimensional profile, the second structure positioned at least partially within the first cylindrical structure, the first cylindrical structure defining at least one detent that is cut at least partially into the wall of the first cylindrical structure and the second structure defining at least one tab protruding radially outwardly from the wall of the second structure that is configured to lock into the at least one detent of the first cylindrical structure to prevent linear movement of the second structure relative to the first cylindrical structure, wherein the second structure is configured to be rotatable relative to the first cylindrical structure for disengaging the at least one tab from the at least one detent for allowing linear movement of the second structure relative to the first cylindrical structure for adjusting an overall length of the cable management device.

22. The cable management assembly of claim 21, wherein the first cylindrical structure defines a plurality of discrete detents that are cut at least partially into the wall of the first cylindrical structure, wherein the at least one tab of the second structure is configured to lock into a selected one of the detents for allowing adjustment of the overall length of the cable management device.

23. The cable management assembly of claim 21, wherein the apertures of the telecommunications frame and the attachment structures of the cable management devices are configured such that the cable management devices can be removably mounted to the telecommunications frame at selected vertical and horizontal locations on the frame.

24. The cable management assembly of claim 21, wherein the second structure is slidable with respect to the first cylindrical structure when the at least one tab is out of the at least one detent for adjusting the length of the cable management device.

25. The cable management assembly of claim 21, wherein the first cylindrical structure defines two sets of detents cut at least partially into opposing sides of the wall of the first cylindrical structure, and the second structure defines two tabs that are protruding from opposing sides of the wall of the second structure, wherein the tabs of the second structure are configured to lock into a selected one of the detent sets for allowing adjustment of the overall length of the cable management device.

26. The cable management assembly of claim 21, wherein one of the first cylindrical structure and the second structure includes a cable retention flange at an end of the cable management device.

27. The cable management assembly of claim 21, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force.

28. The cable management assembly of claim 27, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force that is along a radial direction relative to the walls of the first cylindrical structure and the second structure.

29. A cable management device for mounting to a telecommunications fixture, the cable management device comprising: a first cylindrical structure defining a wall with a circular lateral cross-dimensional profile; and a second structure defining a wall with a circular lateral cross-dimensional profile, the second structure positioned at least partially within the first cylindrical structure; wherein the first cylindrical structure defines at least one detent that is cut at least partially into the wall of the first cylindrical structure and the second structure defines at least one tab protruding radially outwardly from the wall of the second structure that is configured to lock into the at least one detent of the first cylindrical structure to prevent linear movement of the second structure relative to the first cylindrical structure, wherein the second structure is configured to be rotatable relative to the first cylindrical structure for disengaging the at least one tab from the at least one detent for allowing linear movement of the second structure relative to the first cylindrical structure for adjusting an overall length of the cable management device.

30. The cable management device of claim 29, wherein the first cylindrical structure defines a plurality of discrete detents that are cut at least partially into the wall of the first cylindrical structure, wherein the at least one tab of the second structure is configured to lock into a selected one of the detents for allowing adjustment of the overall length of the cable management device.

31. The cable management device of claim 29, wherein the second structure is slidable with respect to the first cylindrical structure when the at least one tab is out of the at least one detent for adjusting the length of the cable management device.

32. The cable management device of claim 29, wherein the first cylindrical structure defines two sets of detents cut at least partially into opposing sides of the wall of the first cylindrical structure, and the second structure defines two tabs that are protruding from opposing sides of the wall of the second structure, wherein the tabs of the second structure are configured to lock into a selected one of the detent sets for allowing adjustment of the overall length of the cable management device.

33. The cable management device of claim 29, wherein one of the first cylindrical structure and the second structure includes a cable retention flange at an end of the cable management device.

34. The cable management device of claim 29, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force.

35. The cable management device of claim 34, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force that is along a radial direction relative to the walls of the first cylindrical structure and the second structure.

36. A cable management device for mounting to a telecommunications fixture, the cable management device comprising: a first cylindrical structure defining a wall with a circular lateral cross-dimensional profile; and a second structure defining a wall with a circular lateral cross-dimensional profile, the second structure positioned at least partially within the first cylindrical structure; wherein one of the first cylindrical structure and the second structure defines a plurality of discrete detents that are cut at least partially into the wall of one of the first cylindrical structure and the second structure, wherein the other of the first cylindrical structure and the second structure defines at least one tab protruding radially from the wall of the other of the first cylindrical structure and the second structure that is configured to lock into a selected one of the discrete detents to prevent linear movement of the second structure relative to the first cylindrical structure, wherein the at least one tab is configured to be locked into a selected one of the discrete detents under a bias force, wherein the second structure is configured to be rotatable relative to the first cylindrical structure for disengaging the at least one tab from the selected detent for allowing linear movement of the second structure relative to the first cylindrical structure for adjusting an overall length of the cable management device.

37. The cable management device of claim 36, wherein the second structure is slidable with respect to the first cylindrical structure when the at least one tab is out of the selected detent for adjusting the length of the cable management device.

38. The cable management device of claim 36, wherein the first cylindrical structure defines the plurality of discrete detents cut at least partially into the wall of the first cylindrical structure, and the second structure defines the at least one tab protruding radially from the wall of the second structure.

39. The cable management device of claim 38, wherein the first cylindrical structure defines two sets of detents cut at least partially into opposing sides of the wall of the first cylindrical structure, and the second structure defines two tabs that are protruding from opposing sides of the wall of the second structure, wherein the tabs of the second structure are configured to lock into a selected one of the detent sets for allowing adjustment of the overall length of the cable management device.

40. The cable management device of claim 36, wherein one of the first cylindrical structure and the second structure includes a cable retention flange at an end of the cable management device.

41. The cable management device of claim 36, wherein the at least one tab is configured to be locked into a selected one of the discrete detents under a bias force that is along a radial direction relative to the walls of the first cylindrical structure and the second structure.
